

Unequal Job Opportunities and Well-Being Inequality: A Latent-Jobs Structural Decomposition

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France: EU-SILC collected in 2016, incomes of 2015, every job priced through EUROMOD.

Discussion draft. The decomposition is preliminary and is not a causal decomposition.

Thank you. This talk asks how much of the inequality in money-metric well-being goes with unequal job opportunities rather than with differences in what people prefer. I will build the answer in four steps: a model of labour supply as choice among latent jobs, which separates preferences from opportunities; two ways of turning choices into euros of well-being; a set of counterfactuals that equalise one source of heterogeneity at a time; and a Shapley rule that allocates the change. The punchline is that the answer depends on the welfare question you ask.

Two households with the same income can be very differently placed.

- Low earnings can mean a taste for short hours, no well-paid job within reach, or household resources that make another arrangement affordable.
- An income distribution records outcomes with these explanations already mixed in.
- In France, well-being inequality differs from consumption inequality:
 - single adults: consumption Gini 0.2633, attained-bundle well-being Gini 0.2498;
 - couples: consumption Gini 0.2268, attained-bundle well-being Gini 0.1974.

Equivalised, household-weighted, within each population. Single-adult and couple households are separate populations and are not compared with each other.

Two people can work the same hours for the same hourly pay and be very differently placed. One chose that job from several that were open; the other took the only offer available. Their incomes are identical and their circumstances are not. The Ginis on the slide make the point in the data: once the leisure a job costs is valued, the dispersion of well-being is not the dispersion of consumption. These are within-sample dispersion comparisons, not welfare-loss estimates. The rest of the talk asks which sources of heterogeneity that well-being inequality goes with.

Each ingredient exists; a decomposition of the level of well-being inequality across them does not.

Labour supply as choice among latent jobs

Aaberge, Dagsvik and Strøm; Dagsvik and Strøm; Dagsvik and Jia; Capéau, Decoster and Dekkers.

Separates tastes from opportunities; used to evaluate *reforms*.

Money-metric welfare with heterogeneous preferences

Fleurbaey and Maniquet; Decoster and Haan; Bargain et al.; Jia and Thoresen; Jacquet, Jia and Thoresen.

A reference encodes a normative position; references sit on deterministic budget sets, or value a *change* between regimes.

Decomposing inequality

Shorrocks; Sastre and Trannoy; Creedy and Hérault; Mühlhan.

Shapley allocation of a *change* between policy, population or time states.

Every ingredient has antecedents, and the contribution is the combination and the empirical answer. The latent-jobs framework, with its excluded opportunity shifters, is due to Aaberge, Dagsvik and Strøm and was set out for applied work by Capéau, Decoster and Dekkers, who already cover singles and couples. Money-metric evaluation with such models exists, but it evaluates a change between policy regimes; our object is a distribution of levels under one policy system. The normative literature following Fleurbaey and Maniquet shows how a reference encodes compensation and responsibility. The closest decompositions, Creedy and Hérault and Mühlhan, decompose a change between situations. The Shapley rule itself is inherited; no new allocation principle is claimed.

Gap: a cross-sectional *level* of well-being inequality, decomposed into preferences and job opportunities inside an estimated latent-jobs model, under welfare measures that can value opportunities themselves.

How much inequality in money-metric well-being is associated with unequal job opportunities rather than heterogeneous preferences, once labour supply is modelled as choice among latent jobs?

1. Do observed labour-supply choices reflect preferences alone, or also heterogeneous latent job opportunities?
2. How do conclusions differ when welfare evaluates the attained bundle versus the ex-ante opportunity prospect?
3. In a restricted P/A/B decomposition, how much welfare inequality is associated with preferences, local access, and earning opportunities?

This is the whole talk on one slide. The first subquestion is behavioural: can we read preferences off choices, or do opportunities get in the way? The second is normative: once we value well-being in euros, does it matter whether we value the job a household ends up in or the prospect it faces? The third is the accounting: inside a restricted decomposition, with preferences, local access and earning opportunities as the three channels, how is inequality allocated? Note the word associated: nothing here is causal.

The latent-jobs model: a job is a package, valued through leisure and consumption.

A package j : employment, one of four occupation groups, weekly hours and an hourly wage, priced through the French tax–benefit system to monthly disposable consumption $C_i(j)$.

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right)$$

L_i value of leisure, shifted by age and, for women, children β_c weight on log consumption, estimated λ_c a units convention

Estimated separately for 1,540 single-adult and 2,223 couple households. The child term is a reduced-form time-constraint shifter, not pure taste.

In words: a household's systematic value of a job is the value of the leisure it leaves plus the log of the consumption it pays. The split matters later, because consumption is observable in euros and leisure is not; it is what turns a utility gap into a proportional consumption adjustment in both welfare measures. For a couple, the leisure term sums both spouses' terms and consumption is joint. The household-specific leisure coefficients, age profiles and the child-related shifter, are the preference channel we later equalise.

A job is chosen because it is valued or because it is plentiful: the choice law carries an opportunity density.

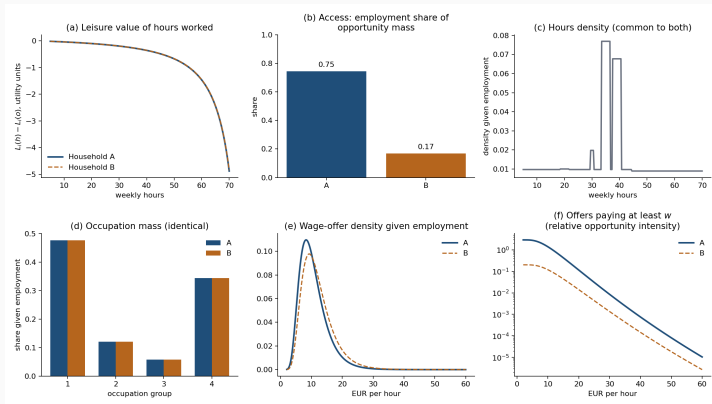
$$P_i(j) = \frac{\exp\{v_i(j)\} g_i(j)}{\int \exp\{v_i(k)\} g_i(k) d\nu(k)}$$

g_i : how intensely package j is available to household i before any choice. It has four blocks: **access** (local unemployment exposure, region, urban or rural location, year), hours bands, occupation, and **wage offers** (education, experience).

Separation rests on maintained restrictions: smooth preferences against banded hours opportunities; local shifters excluded from preferences; wage offers independent of hours given occupation.

In words: the probability of being seen in a job is its value times its availability, normalised over all jobs. If everyone faced the same density, we would be back to a standard model where every difference in behaviour must be a difference in taste. Because the density is household-specific, part of behaviour can be attributed to reachability. A spike at a band of hours is read as availability, not a kink in tastes; the local unemployment measure and the geography shift access and appear nowhere in utility. These are maintained restrictions, not tested ones, and the regional variation is not a causal design.

What unequal opportunity means here: two matched men with almost identical tastes.



Same occupation group, hours band and wage quintile. A has 14.5 times B's employment mass; B's wage-offer location is 11.5 log points higher. A stated-rule teaching example: not causal, not representative.

These two single men look alike in an income table and have nearly identical estimated leisure profiles. If both were indifferent between all jobs and not working, employment would be 0.75 of A's opportunity mass and 0.17 of B's; the gap comes from local unemployment exposure and region. B's wage offers are better, yet A faces more offers paying at least any given wage over essentially all offer mass. Neither is better placed on every margin, and which one is better off depends on how welfare weighs the number of reachable jobs against what the realised job pays. That is exactly the question that separates the two welfare perspectives.

First subquestion: choices reflect opportunities as well as preferences.

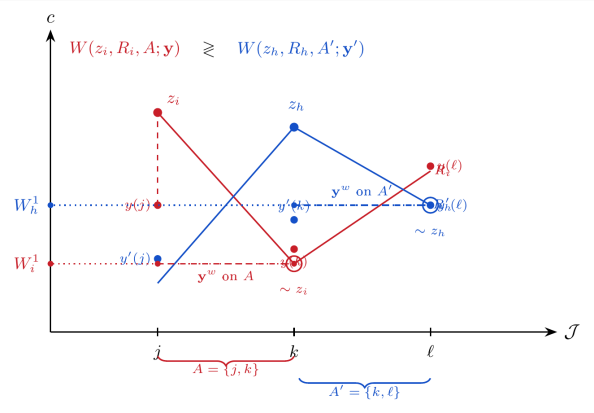
Single adults	free coordinates	population fit (mean absolute deviation)
Latent jobs, household-specific opportunities	41	0.0140
Common opportunity distribution	10	0.0273
Common opportunity shape; employment and hours moved into utility	16	0.0264

Both re-estimated common-opportunity benchmarks are worse by at least 141.64 on the same criterion, households and sampled alternatives; the occupation margins deteriorate by an order of magnitude.

Not converted into a formal test. A better criterion does not by itself establish that a mechanism has been identified.

The first subquestion. Take away the household-specific opportunity block, re-estimate everything else, and the model does worse, on the same households and the same criterion. The deterioration is concentrated where the opportunity block does its work: population fit roughly doubles its error and occupation margins deteriorate most. I do not turn this into a likelihood-ratio test, because the sampled-alternative criterion is not the likelihood of the observed data. Fit is not perfect either: the observed concentration at exactly thirty-seven hours is underpredicted in every group, and extensive-margin accuracy for men is withheld because integration error is too large.

From choices to well-being: give every job the same consumption and ask what is as good as the actual situation.



Own-set equal-consumption equivalents: the theoretical construction from the companion theory paper. Two individuals, with preferences R_i and R_h and ability sets $A = \{j, k\}$ and $A' = \{k, \ell\}$, attain bundles z_i and z_h . For each individual, a common consumption level is assigned to every job in their own set; the level at which the preferred reference job becomes indifferent to the attained bundle is that individual's money metric, W_i^1 and W_h^1 . The two metrics are then directly comparable. Adapted from Haydar and Maniquet (2026), work in progress. This is the deterministic construction; the estimated ex-ante measure is its extension, defined in Section 3.

Estimated utility is not an interpersonal welfare index: preferences differ, so the same utility number means different things for different people. Both money metrics descend from this construction. Each individual is evaluated with their own preferences and their own set of jobs. Give every job in that set the same consumption; the individual then prefers one of them, and the consumption at which that job is exactly as good as the attained bundle is their money metric. The point is that these metrics are comparable across people whose preferences and job sets differ. The figure is a theoretical illustration with no estimated value in it. The choice of reference is the welfare question, and the next two slides ask two.

EA, the opportunity-prospect perspective: what flat consumption makes the whole job prospect equally valuable?

$$J_i = \int e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j), \quad H_i = \int e^{L_i(j)} \widehat{g}_i(j) d\nu(j),$$
$$M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

- J_i : the actual prospect; every job enters, weighted by how available it is and how much it is valued.
- H_i : the same jobs, availability and preferences, with pay differences removed.
- Opportunities enter welfare directly: access moves which jobs carry weight; earning opportunities move what they pay.

The measure treats taste-shock variety as part of what makes a rich prospect valuable: a normative position, not a consequence of estimation.

In words: EA is the constant monthly consumption at which a prospect with the household's own jobs and availability, but equal pay everywhere, is worth exactly as much as the prospect it actually faces. A common rescaling of the opportunity density cancels, so what matters is how opportunity weight is spread across jobs. I show this perspective first because this talk is about opportunities; that is an order of presentation, not a ranking. Under the extreme-value shocks, $\log J$ is the expected utility of the best reachable job up to a constant, which is why taste-shock variety counts; I flag that as a normative choice. The ex-ante calculation passed its numerical checks, which certifies the computation, not the model's assumptions.

ATT, the attained-bundle benchmark: what consumption without work is as good as the job actually held?

$$M_i^{\text{att}} = C_i^{\text{obs}} \exp \left\{ \frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right\}$$

- Reference: non-employment o , which every household can reach.
- Opportunities matter only through which job is attained and what it pays; jobs not taken play no role.
- For non-workers $M_i^{\text{att}} = C_i^{\text{obs}}$; for workers it lies below observed consumption by the value of the leisure given up.

On the current empirical domain the equal-consumption reference job is non-employment for every household: a verified property of the estimated specification, not a general theorem.

In words: ATT is observed consumption, adjusted by the leisure the job costs relative to not working, converted at the rate one over the consumption weight. The general construction evaluates the attained bundle against the job the household would most prefer if all jobs paid the same; here that job is non-employment for everyone, so the measure coincides with the staying-home equivalent. Once the job is fixed, the opportunity density has no direct term. This is the natural benchmark: it values the outcome reached.

The two measures answer different welfare questions.

	EA: the opportunity prospect	ATT: the attained bundle
Question	How valuable is the distribution of job prospects the household faces?	How well off is the household in the bundle it actually attains?
Reference	The same opportunity environment, with consumption equalised across jobs	A common non-employment state
How opportunities enter	Directly, as weights on every job	Only through the realised job

Neither perspective corrects the other, and neither is designated primary. The choice between them is a substantive normative decision that is not made here.

This is the hinge of the talk. The attained-bundle measure evaluates the bundle eventually reached against a common non-employment reference. The ex-ante measure evaluates the household’s entire distribution of potential job outcomes against a reference prospect that keeps the same opportunity environment and equalises consumption across jobs. Because opportunities reach them by different routes, the two can attribute inequality to different labour-market channels. When they do, the difference tells us what each welfare question makes visible. It is not a disagreement to be resolved by choosing a winner.

The decomposition equalises three channels and allocates the change in the Gini over all orders.

P preferences: age profiles and the child-related shifter.

A local access: local unemployment exposure, region, urban or rural location, year. Not total opportunity.

B earning opportunities: systematic wage-offer location, not wage dispersion.

Held fixed: household resources, needs and composition, in every coalition, with no allocated share.

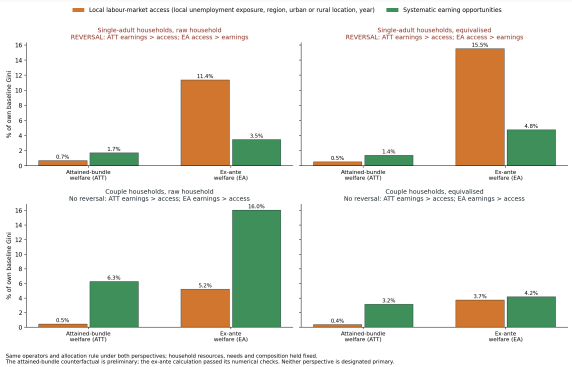
$$\phi_k^p = \sum_{S \subseteq \{P, A, B\} \setminus \{k\}} \frac{|S|! (3 - |S| - 1)!}{3!} \left[v^p(S \cup \{k\}) - v^p(S) \right]$$

$v^p(S) = I^p(\emptyset) - I^p(S)$: the fall in the household-weighted Gini of welfare M^p , $p \in \{\text{EA}, \text{att}\}$, when the channels in S are set to a common reference.

Household resources, needs and composition are held fixed in the current decomposition. The access channel is local access, not total opportunity. The decomposition is preliminary and makes no causal claim.

In words: a channel's Shapley contribution is its marginal effect on inequality averaged over every order in which the three equalisations can be introduced, and the three contributions add up exactly to the total change. Each coalition is a full structural counterfactual: coefficients are never changed, welfare is recomputed for every household under each perspective, and inequality is measured again. The same operators and rule are applied to both perspectives, which is what makes their orderings comparable. Contributions are signed: a negative term means equalising that channel raises inequality on average over orders. Adding resources, needs and composition as a fourth channel is a planned extension; with it, access plus earnings would still be the opportunity component, and the broader sum would not be an opportunity share.

Central result: earning opportunities dominate under ATT; access dominates under EA for single adults; couples show no reversal.



This is the result. Read each panel as the same two operators, local access and earning opportunities, evaluated under two welfare questions. For single adults, earning opportunities are the larger channel when we value the attained bundle, and access is the larger channel when we value the prospect, on both reporting scales. For couples, earning opportunities stay in front under both. The mechanism is the route by which opportunities enter: the attained-bundle measure sees only the realised job, where the wage drives consumption; the ex-ante measure weights every job by its availability, so the access environment enters directly. Why single adults reverse and couples do not is not tested here; one reading is that a single adult’s participation rests on one earner’s access. The comparison is also not a pure contrast of definitions, because the two integrations use different designs.

Access and earning-opportunity contributions as a percentage of each perspective’s own baseline Gini. Access is local access, not total opportunity; household resources, needs and composition are held fixed. Neither perspective is designated primary.

EA: access and earnings account for 7.9–21.3% of baseline inequality; for single adults access is about three times earnings.

	scale	access Gini points	earnings Gini points	access + earnings % of baseline	ordering
Single adults	raw	0.01812	0.00555	14.8	access > earnings
	equalised	0.02183	0.00673	20.3	access > earnings
Couples	raw	0.00667	0.02049	21.3	earnings > access
	equalised	0.00449	0.00504	7.9	earnings > access

Single adults: access is 3.3 times earnings before equalisation and 3.2 times after it.

Household resources, needs and composition held fixed; access is local access. For couples, equalisation moves the share from 21.3% to 7.9% and turns the preference term negative, so no directional claim about preferences is made.

The ex-ante numbers. The calculation passed its numerical checks, including an independent implementation. For couples the scale convention is substantively consequential: the preference contribution goes from 0.00291 Gini points to −0.00212 when we equalise. A negative Shapley term means that, averaged over orders, equalising preferences raises inequality; it does not show that preference heterogeneity is equalising in any causal or welfare sense. That is why both scales are always shown. These are restricted accounting shares, not the total share of well-being inequality caused by unequal job opportunities.

ATT benchmark: access and earnings account for 1.9–6.7% of baseline inequality, and earning opportunities are larger throughout.

		scale	access	earnings	access + earnings	ordering
			Gini points	Gini points	% of baseline	
Single adults	raw		0.0016	0.0040	2.4	earnings > access
	equivalised		0.0013	0.0034	1.9	earnings > access
Couples	raw		0.0009	0.0128	6.7	earnings > access
	equivalised		0.0007	0.0062	3.5	earnings > access

Preliminary: the attained-bundle counterfactual integration is still being validated and covers short hours sparsely. The preference contribution changes sign with equivalisation in both samples, so no directional claim is made about it. Household resources, needs and composition held fixed.

The attained-bundle benchmark. Earning opportunities are larger than local access for both household types and on both scales. Every coalition Gini and contribution reproduces closely under an independent simulation run, and the allocation closes to numerical precision; those checks validate the calculation for the stated game, not its maintained assumptions. The lower part of the hours range is represented sparsely in the integration sample, and the effect of that on this attribution has not been fully quantified, which is why this exercise stays explicitly preliminary.

What these results are not.

- **Preliminary.** A restricted structural accounting exercise; the attained-bundle counterfactual integration is still being validated.
- **No causal claim.** Separation rests on maintained functional-form and exclusion restrictions; no causal effect of geography, education or occupation is estimated.
- **No parameter uncertainty yet.** Estimation uncertainty is not propagated through either decomposition; repeat runs are simulation checks, not confidence intervals.
- **Preferences are not responsibility.** The preference channel is systematic utility heterogeneity, including a reduced-form child-related shifter; it is not equated with what households are responsible for.
- **Held fixed and narrow.** Household resources, needs and composition are held fixed; access is local access, not total opportunity. The shares are not the total share of inequality due to unequal job opportunities.

Five boundaries, all on the slide on purpose. The decomposition is preliminary. Nothing is causal: the regional variation supports the access block only conditional on the maintained restrictions. Parameter uncertainty has not been propagated, so there are no confidence intervals on any share. The preference channel is not a responsibility channel: it contains a reduced-form child term that can capture childcare constraints as well as tastes, and deciding what people should be held responsible for is a normative step this paper does not take. Finally, the exercise holds household resources, needs and composition fixed and equalises only local access and wage-offer location, so the percentages are neither a complete decomposition nor an estimate of everything unequal opportunity does. Equivalisation is a convention and changes the couples results, which is why both scales are shown.

Conclusion: which labour-market inequality matters depends on the welfare question.

1. **Choices reflect opportunities, not preferences alone.** Common-opportunity benchmarks fit worse on the same households.
2. **The welfare question changes the answer.** Under **EA**, access and earnings account for 7.9–21.3% of baseline inequality and access leads for single adults; under **ATT**, 1.9–6.7% and earnings lead. Couples show no reversal.
3. **In the restricted P/A/B decomposition**, local access and earning opportunities are a bounded part of well-being inequality; the preference contribution is not sign-robust to equivalisation.

Next: household resources, needs and composition as a fourth channel; parameter uncertainty; completing validation of the attained-bundle integration.

To close, the three answers. Observed choices carry information about opportunities, and a model that gives everyone the same jobs fits worse. Once choices become well-being, the importance assigned to different labour-market inequalities depends on whether welfare evaluates the realised outcome or the opportunity prospect itself: earning opportunities lead when we value the attained bundle, access leads for single adults when we value the prospect, and couples do not reverse. And within the restricted decomposition, with resources, needs and composition held fixed, these channels are a bounded share of inequality. Neither perspective is designated primary. Thank you.